

Noorinder

Error Log for FindAndReplace

1. Incorrect File Reading Method

- Error: Used `Files.readString()` instead of `Files.readAllBytes()`.
- Cause: `Files.readString()` is less compatible and requires additional handling for encoding.
- Fix: Replaced `Files.readString()` with `Files.readAllBytes()` to ensure broader compatibility.

Error Code:

```
String content = Files.readString(Paths.get(filename));
```

Fixed Code:

```
String content = new String(Files.readAllBytes(Paths.get(filename)));
```

2. Case-Sensitive Replacement Logic

- Error: Replacement logic does not account for case variations of the search word.
- Cause: The `.replace()` method is case-sensitive by default.
- Fix: Used regex-based replacement with the `(?i)` flag for case insensitivity.

Error Code:

```
String updatedContent = content.replace(searchWord, replacementWord);
```

Fixed Code:

```
String updatedContent = content.replaceAll("(?i)" + searchWord, replacementWord);
```

3. File Overwriting Without Confirmation

- Error: The program overwrites the file directly without user consent.
- Cause: There is no prompt for confirmation before overwriting the file.
- Fix: Added a prompt to confirm overwriting.

Error Code:

```
try (PrintWriter writer = new PrintWriter(filename)) {  
    writer.write(updatedContent);  
}
```

Fixed Code:

```
System.out.print("The file will be overwritten. Do you want to proceed? (yes/no): ");  
String confirmation = scanner.nextLine();  
if (confirmation.equalsIgnoreCase("yes")) {  
    try (PrintWriter writer = new PrintWriter(filename)) {  
        writer.write(updatedContent);  
    }  
    System.out.println("File has been updated.");  
} else {  
    System.out.println("Operation canceled.");  
}
```

4. Misleading File-Not-Found Error Message

- Error: The file-not-found message does not mention the actual filename.
- Cause: The error message is too generic, making it unclear which file is missing.
- Fix: Updated the error message to include the filename.

Error Code:

```
System.out.println("File not found. Please ensure the file path is correct.");
```

Fixed Code:

```
System.out.println("The file '" + filename + "' was not found. Please check the file path and try again.");
```